Host Utilities

TATA POWER

TATA POWER-DDL
Towards a Greener Tomorrow

BSES Rajdhani Power Limited

BSES Yamuna Power Limited

NPCL
NOIDA POWER COMPANY LIMITED

Knowledge Partner

Power Foundation of India

SESSION PARTNER

DEPARTMENT OF SCIENCE & TECHNOLOGY

ORGANIZER

ISGF
India Smart Grid Forum

Supporting Ministries

MINISTRY OF POWER
GOVERNMENT OF INDIA

NITI Aayog

MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

DEPARTMENT OF SCIENCE & TECHNOLOGY

AIDA
Collaboration | Innovation | Excellence

India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

SESSION : NEW TECHNOLOGIES: AI–ML–VR–AR ROBOTICS & APPLICATIONS & RELIABLE COMMUNICATIONS FOR UTILITIES

Staining and reliable communication in AMI implementations

Presented By

Sandeep Dhamija, CEO , TP Power Plus Ltd.

Introduction – IoT based Ecosystem & Communication Challenge

India SMART UTILITY Week 2026 | ISGF India Smart Grid Forum

IoT Ecosystem in Indian Power Distribution

- With large scale AMI adaption, Indian distribution systems now operate with **millions of IoT devices**
- These devices uses **RF mesh and Cellular Networks** for communication with cellular leading the way.
- The data enables key utility processes such as **billing, energy audit, revenue protection, demand management and various other key business analytics**.

Key Challenge – Communication Reliability

- Large-scale IoT ecosystems depend on **reliable communication from millions of devices**.
- **Poor communication results in:**
  - Issue with billing, mainly concept of prepaid billing
  - Inaccurate analytics and energy accounting
  - Reduced operational efficiency and monitoring capability
  - **Underutilization of smart metering investments**

Reliable communication is critical for realizing the full value of large-scale smart metering deployments.

Context – Communication Management at Scale

1. Large scale deployment of IoT devices involve **management of millions of communication nodes.**

2. Scale fundamentally changes the approach for communication management

3. Traditional Approach Limitations

- Manual monitoring and root-cause analysis on basis of field visits.
- Suitable for **SCADA / DT Automation (limited nodes)**
- **Not feasible for AMI scale**

4. Need for Intelligent Systems

- **AI-driven analytics for communication health monitoring**
- **Effective Communication Management has to deal with Meter, NIC, SIM, IP address,** Cloud, HES, integration
- Automated identification of probable communication issues and implementing strategies for maintaining reliable and sustaining communication

Reliable communication ensures effectiveness of AMI system

Relevance – Indian Smart Metering Scenario

India is undertaking **large-scale smart metering deployment under RDSS**

Target: Installation of over **200 million Smart Meters in near future**

Lessons from Past Initiatives:

- Earlier program's such as **R-APDRP and IPDS** introduced **AMR-based systems**
- Over time, many systems became **ineffective due to poor sustainability of communication performance**

Critical Concern: If **communication reliability is not maintained**, the significant investment in smart metering infrastructure may **not deliver the expected operational and financial benefits**

Technology deployment without strong communication management systems may lead to limited operational benefits

1. Pre-Installation Checks (Preventive Approach)

India SMART UTILITY Week 2026 | ISGF India Smart Grid Forum

Common issues during installation:

- NIC / Communication module issues
- SIM activation/failures
- Incorrect IP / APN configuration
- Incorrect mapping of meter-sim-IP address

Stakeholder Integration Layer

Meter Manufacturer(Stake Holder-1):

- Manufacturing of meters with assembly of NIC & SIM

SIM Supplier (Stake Holder-2):

- SIM supplier deliver the SIM's
- IP assignment
- Activation in Test state valid for 3 months, becomes non-communicating if not made "Active"

Meter Onboarding on HES(Stake Holder-3):

- Process can be initialized when the Meter No., SIM no., IP address are mapped in system

Innovation

First of its kind where Comprehensive 100% Communication testing of each meter before dispatch is performed successfully

Cost-Effective Test Bench Development to match scale of testing

- Inhouse Designed & developed the test benches
- Design the meter charging durable pins for quick operation
- Testing of 60 meters in a one bench with an end-to-end cycle time of 30 mins

Automation & Mobile Application

- Automated testing via in-house mobile app; reduced manual intervention and optimized manpower.

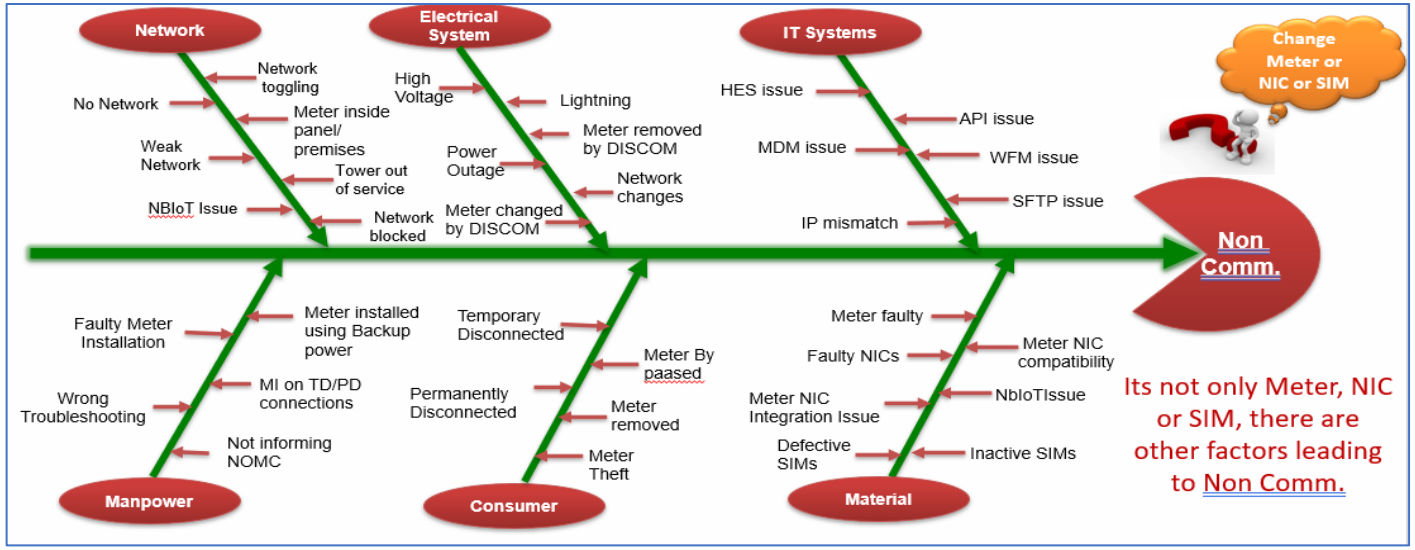
2. Post-Installation Communication Sustainability

Challenges in maintaining Communication Success Rate:

- **Communication Success Rate** is the most critical KPI in smart metering; however, non-communicating meters are often addressed through hardware replacement rather than root-cause analysis.

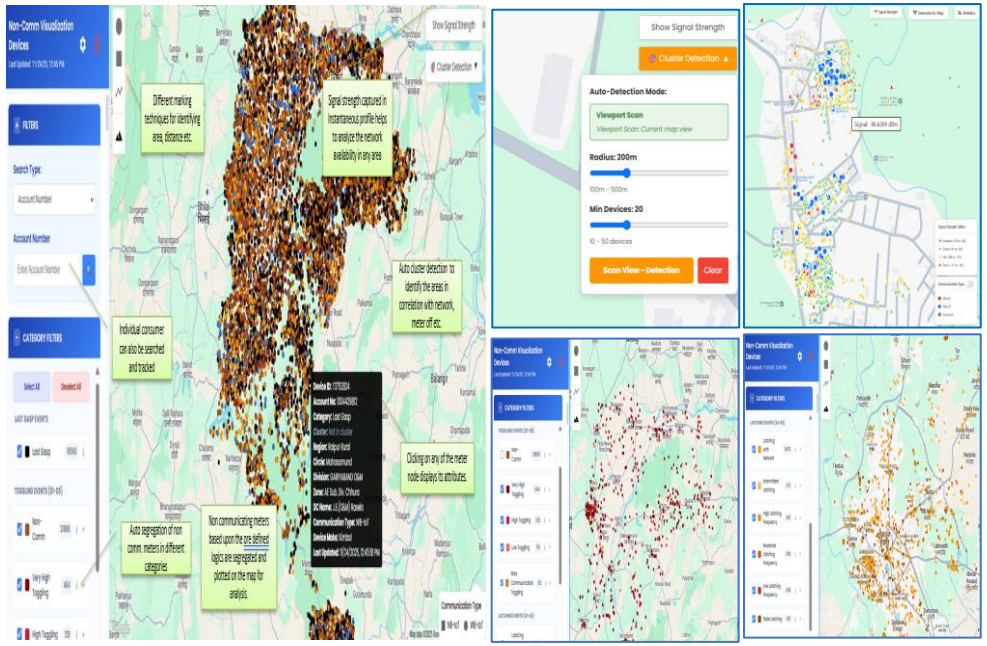
- **Manual analysis** is inadequate to evaluate the multiple factors affecting communication across millions of meters.

Root Cause Identified



Innovation

- This project introduces a **first-of-its-kind AI-based Communication Management System** that automatically identifies root causes of communication failures and provides **actionable insights to improve efficiency and communication reliability**.
- System predicts reasons of non communication, Detects clusters and provides better visual analytics



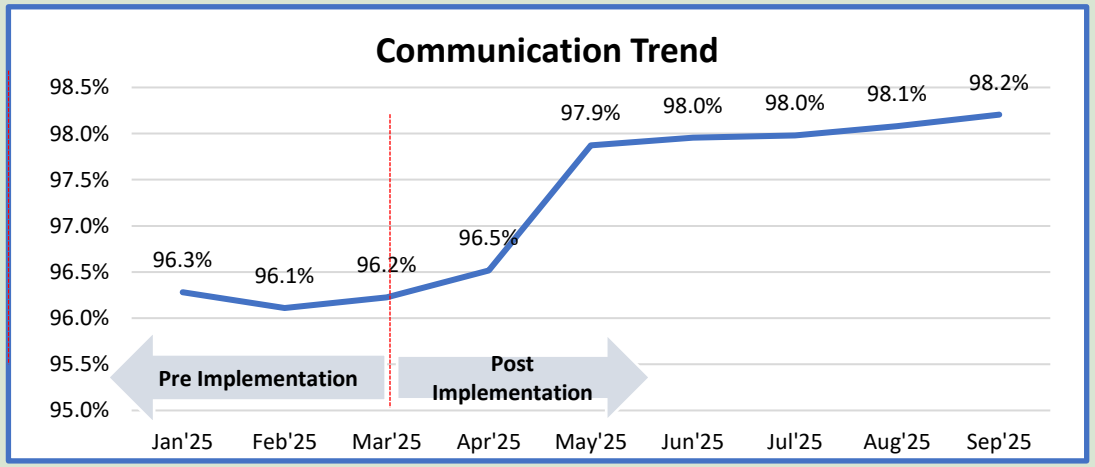
USE CASE / CASE STUDY

Pre-Deployment Validation

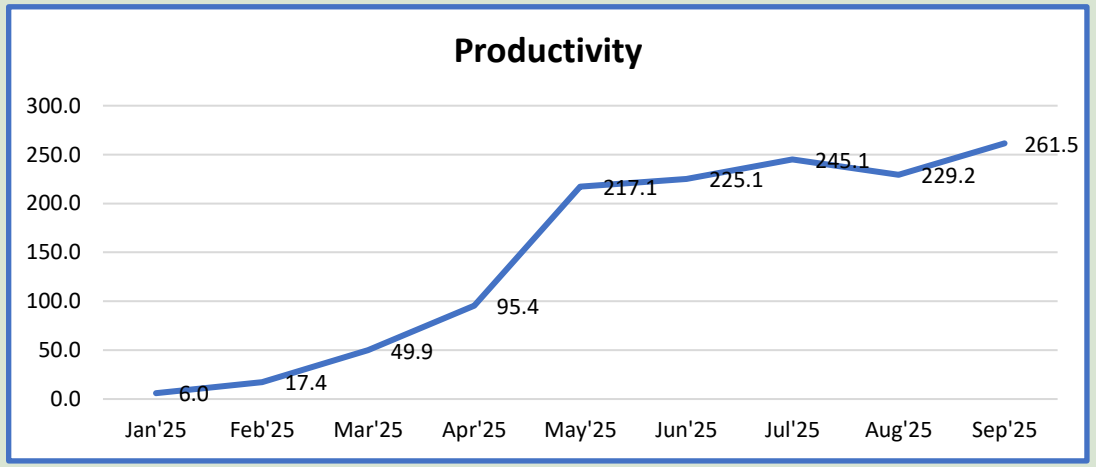
1. **Achieved 6,700 meters/day communication testing**
2. During pre-deployment testing, approximately **52,000+ meters were identified as 'FAILED'** and held back at the warehouse for troubleshooting.
3. **52,000 meters prior to installation lead to direct cost saving of INR 10.4 million**

Non-Communication Visualization Tool

1. The impact post implementation is already visible on the communication success rate and productivity of the field staff.



% is calculated excluding field exceptions not pertaining to AMISP



Avg. count of sites attended per person per month

2. These tools helped in better SLA management of around INR 400 million over 10 years project period
3. Better Communication of meters would help utilities in better Operational Mamagement

KEY TAKEAWAYS / RECOMMENDATIONS

1. Communication is the Backbone of AMI systems

- Meter installation alone does not ensure success
- Reliable communication is essential for **data-driven utility operations**

2. Adopt AI & Automation

- Use **advanced analytics tools** to monitor millions of nodes
- Enable **issue detection and prioritization with communication sustenance strategies**

3. Continuous Performance Monitoring

- Track **communication performance trends**
- Identify early signs of **network degradation**
- Use analytics to guide **preventive & corrective actions**

Host Utilities

TATA POWER

TATA POWER-DDL
Towards a Greener Tomorrow

BSES
BSES Rajdhani Power Limited

BSES
BSES Yamuna Power Limited

NPCL
NOIDA POWER COMPANY LIMITED

Knowledge Partner

Power Foundation of India

SESSION PARTNER

DEPARTMENT OF SCIENCE & TECHNOLOGY

ORGANIZER

ISGF
India Smart Grid Forum

India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

Supporting Ministries

MINISTRY OF POWER
GOVERNMENT OF INDIA

NITI Aayog

MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

DEPARTMENT OF SCIENCE & TECHNOLOGY

AIDA
Collaboration | Innovation | Excellence

THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

www.isuw.in